

1. Administrative & Study Identification

Full Study Title: A Study of Enlicitide Decanoate (MK-0616, an Oral PCSK9 Inhibitor) in Children and Adolescents With Heterozygous Familial Hypercholesterolemia (MK-0616-029) **Short Title/Acronym:** CORALreef Pediatric (MK-0616-029) **Protocol Number:** MK-0616-029 (EU CT: 2024-519068-42-00) **NCT Number:** NCT07058077 **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) **Investigational Product:** Enlicitide Decanoate (MK-0616) **Phase of Development:** Phase 2/3 **Medical Monitor:** [Merck Sharp & Dohme Clinical Director / Medical Monitor]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether enlicitide decanoate is safe and effective in lowering low-density lipoprotein cholesterol (LDL-C) levels more than a placebo in children and adolescents with Heterozygous Familial Hypercholesterolemia (HeFH). **Secondary Objectives:** To evaluate the overall safety, tolerability, and pharmacokinetics of enlicitide decanoate over time in a pediatric population.

2.2 Background & Rationale Heterozygous Familial Hypercholesterolemia (HeFH) is a genetic disorder resulting in elevated levels of LDL-C in the blood, which significantly increases the risk of early-onset atherosclerotic cardiovascular disease (ASCVD). Despite current standard-of-care lipid-lowering therapies (LLTs) such as statins, many patients do not reach target LDL-C levels. This study aims to address the unmet medical need for a once-daily, orally administered PCSK9 inhibitor to effectively and safely manage high cholesterol in pediatric and adolescent patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (in Part B) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: [Approximately 100-500 participants, typical for Phase 2/3 pediatric stages] **Number of Sites:** 19 locations **Estimated Study Start:** 21-Aug-2025 **Estimated Primary Completion:** 04-Dec-2033

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 6 to 17 years. 2. Possible or definite diagnosis of HeFH based on a locally accepted diagnostic algorithm or diagnosis by genetic testing results. 3. Fasted LDL-C value (evaluated by the central laboratory) that is ≥ 130 mg/dL. 4. Receiving either an optimized daily dose of statin ($\pm$ nonstatin LLT); or a nonstatin LLT with documented intolerance to at least 2 different statins or refusal of statin therapy. Must be on a stable dose of all background LLTs for at least 30 days prior to screening.

4.2 Exclusion Criteria 1. History of homozygous FH based on genetic or clinical criteria, or history of known compound heterozygous FH, or double heterozygous FH. 2. History of nephrotic syndrome. 3. Any clinically significant malabsorption condition based on principal investigator assessment. 4. Previous or current treatment with other cholesterol-lowering medications, including injectable PCSK9 inhibitors, without an adequate washout period.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Part A:** Enlicitide decanoate orally once daily (QD) at a dosage determined by age for up to 2 weeks. * **Part B:** Enlicitide decanoate or Placebo orally QD at a dosage determined by age for up to 24 weeks. * **Open-Label Extension:** Enlicitide decanoate for participants who complete Part A or B. **Route of Administration:** Oral (daily pill) **Duration of Treatment:** Up to 24 weeks for the main placebo-controlled efficacy phase (Part B), followed by an Open-Label Extension period.

5.2 Concomitant & Prohibited Therapy Permitted: Optimized and stable daily doses of statins and non-statin lipid-lowering therapies (LLT). **Prohibited:** Concurrent use of other PCSK9 inhibitors or unapproved lipid-lowering investigational agents without adequate washout.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to 0	Informed Consent/ Assent, Medical History, Fasting Lipid Panel (Central Lab)
Baseline	Day 0	Randomization, First Dose (Part A/B), PK Baseline
Interim	Weeks [X]	Safety Labs, PK/Tolerability Assessments, Efficacy Assessment (LDL-C)
End of Study (Part B)	Week 24	Final Efficacy Assessment, IP Accountability, Transition to Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percent change from baseline in LDL-C levels at Week 24. **Measurement Method:** Central Laboratory Analysis of fasting lipid panels.

7.2 Secondary & Exploratory Endpoints 1. Safety and tolerability profile of enlicitide decanoate in children and adolescents. 2. Pharmacokinetics of enlicitide decanoate over time in a pediatric body.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via investigator interviews, laboratory results (including liver function and muscle enzymes), and physical examinations. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent external committee details for continuous safety monitoring in the pediatric population.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcome evaluation. Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Powered appropriately to detect a statistically significant reduction in LDL-C compared to placebo at Week 24 ($\alpha = 0.05$). **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) or Multiple Imputation suitable for long-term lipid-lowering trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:** Validated eCRF locking mechanisms and rigorous central laboratory data transfer tracking.

11. Study Identifiers & Governance

Primary ID: {{MK-0616-029}} **Registry Number:** {{NCT07058077}}
Sponsor/Lead Organization: {{Merck Sharp & Dohme LLC}} **Study Title:** {{CORALreef Pediatric}} **Official Regulatory Title:** {{A Study of Enlicitide Decanoate (MK-0616, an Oral PCSK9 Inhibitor) in Children and Adolescents With Heterozygous Familial Hypercholesterolemia (MK-0616-029)}}

12. Clinical Framework (HeFH Specific)

Primary Condition: {{Heterozygous Familial Hypercholesterolemia (HeFH)}} **Diagnosis Threshold:** {{Fasted LDL-C $\geq$ 130 mg/dL}} **Medical Methodology:** {{Standard Background LLT (Statins) + Oral PCSK9 Inhibition}} **Optimization Target:** {{Clinically meaningful percentage reduction in LDL-C}}

13. Study Procedures & Methodology

Management Pathway: {{Pediatric HeFH lipid-lowering clinical pathway}} **Education Protocol (HCP):** {{Study protocol training, pediatric dosing algorithms, safety monitoring}} **Education Protocol (Patient):** {{Medication adherence, pill-swallowing capability, lifestyle/dietary maintenance}} **Quality Audit Mechanism:** {{Central laboratory consistency tracking & site compliance monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 24 weeks (Part B) + Open-Label Extension}} **Onsite Visit Cadence:** {{Screening, Baseline, and protocol-defined regular intervals up to Week 24}} **Remote Monitoring Frequency:** {{As specified by clinical monitoring plan}} **Checklist Review Interval:** {{Prior to Part A/B transition and Open-Label Extension enrollment}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				